

Agentic AI Social Media Generator

Task 2 Internship Project Report | Local Gemma 3:4B via Ollama

1. Executive Summary

This project implements a fully autonomous, offline Agentic AI Social Media Content Generator built inside the `agentic_ai_task_2` project workspace. Utilizing Google's Gemma 3:4B LLM running locally through Ollama, the agent plans, generates, reviews, and persists multi-platform content tailored for LinkedIn, Instagram, X/Twitter, and YouTube Shorts.

2. Agent Architecture & Tool Suite

The agent architecture comprises 6 dedicated tools orchestrated sequentially:

- 1. Trend Idea Generator** - Generates 3 content angles locally and selects the optimal approach.
- 2. Caption Writer** - Drafts scroll-stopping hooks, main post body, and strong call-to-actions.
- 3. Hashtag Generator** - Produces target platform hashtags returned as clean Python lists.
- 4. Reel Script Generator** - Creates structured 30-second video scripts with visual and audio cues.
- 5. Content Reviewer** - Evaluates content quality, platform alignment, and improvement areas.
- 6. File Saver (Mandatory)** - Persists outputs as structured Markdown files in designated output paths.

3. Local JSON Persistent Memory System

A dedicated `MemoryManager` module persists user queries, selected content angles, hooks, and saved file paths to `outputs/memory.json`. Before generating new content, the agent queries previous records to maintain contextual continuity and style alignment across sessions.

4. Verification & Test Suite Results

All 10 required test cases were executed end-to-end against local Gemma 3:4B:

#	Topic	Platform	Output Type	Status
TC-01	YOLOv8 helmet detection	LinkedIn	Post + Reel Script	PASSED
TC-02	AI internship experience	LinkedIn	Post	PASSED
TC-03	Python learning journey	Instagram	Reel Script	PASSED
TC-04	Machine learning project	X/Twitter	Post	PASSED
TC-05	Data annotation struggles	LinkedIn	Post	PASSED
TC-06	Model failed in production	Instagram	Reel Script	PASSED
TC-07	Final year project	LinkedIn	Post	PASSED
TC-08	Computer vision project	YouTube Shorts	Reel Script	PASSED
TC-09	Debugging FastAPI	X/Twitter	Post	PASSED
TC-10	Open-source contribution	LinkedIn	Post	PASSED

5. Conclusion

The project successfully meets all internship requirements, operating 100% locally with zero paid API dependencies. All output paths (outputs/generated_posts/, outputs/generated_scripts/, outputs/saved_results/) have been verified on Windows.